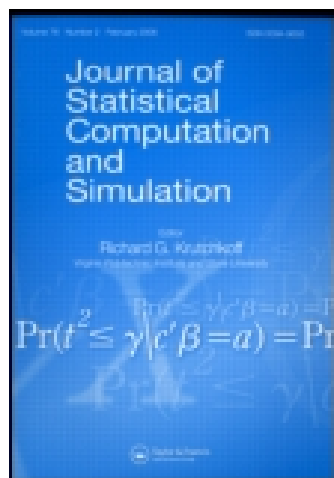


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Journal of Statistical Computation and Simulation

Publication details, including instructions for authors and subscription information:

<http://www.tandfonline.com/loi/gscs20>

Statistical analysis of controlled calibration model with replicates

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Published online: 20 Dec 2011.

To cite this article: Betsabé Blas, Heleno Bolfarine & Victor. H. Lachos (2013) Statistical analysis of controlled calibration model with replicates, Journal of Statistical Computation and Simulation, 83:5, 941-961, DOI: [10.1080/00949655.2011.643410](https://doi.org/10.1080/00949655.2011.643410)

To link to this article: <http://dx.doi.org/10.1080/00949655.2011.643410>

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Statistical analysis of controlled calibration model with replicates

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(Received 3 January 2011; final version received 19 November 2011)

In this work, we generalize the controlled calibration model by assuming replication on both variables. Likelihood-based methodology is used to estimate the model parameters and the Fisher information matrix is used to construct confidence intervals for the unknown value of the regressor variable. Further, we study the local influence diagnostic method which is based on the conditional expectation of the complete-data log-likelihood function related to the EM algorithm. Some useful perturbation schemes are discussed. A simulation study is carried out to assess the effect of the measurement error on the estimation of the parameter of interest. This new approach is illustrated with a real data set.

Keywords: local influence; regression model; linear calibration model; measurement error model; Berkson model

1. Introduction

In analytical chemistry, a calibration model is a general method for determining the concentration of a substance in a sample. For instance, in Section 5, concentration of barium element in a sample from fitting a calibration model, which encompasses two stages is obtained. In the first stage, standard solutions with known concentrations (say, X) chosen to cover a required range were prepared, and the absorbance (say, Y) of each solution was measured. In the second step, the absorbance (say, Y_0) of the unknown concentration (say, X_0) solution was measured.

In order to have sufficient information related to the first step from the calibration model, a minimum of four standard concentration points are prepared. The estimates is poor if the number of points in a calibration curve is small; the accuracy of the estimates increases if the number of data points increases, and preparing such a large number of standard solutions is time-consuming and expensive.

Usually, two or three replicates of the standard solutions are prepared, although ISO 8466 recommends a minimum of 10 [1]. Replicates are obtained from different standard solutions prepared independently. Replicates could also be obtained from different injections of each standard solution, and the number of experimental measures increases considerably. This information can be added into the calibration model, as discussed below.

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In the first stage or calibration curve, if the measurement in all the standard concentrations is replicated, one observes the measurements $(X_i, Y_{ij})(j = 1, \dots, m_i \text{ and } i = 1, \dots, n)$ such that

$$Y_{ij} = \alpha + \beta X_i + \epsilon_{ij}, \quad j = 1, \dots, m_i \text{ and } i = 1, \dots, n, \quad (1)$$

where ϵ_{ij} are i.i.d. $N(0, \sigma_\epsilon^2)$.

In the second stage or prediction stage, one observes Y_{0i} , $i = 1, \dots, r$, on a fixed unknown value X_0 such that

$$Y_{0i} = \alpha + \beta X_0 + \epsilon_{0i}, \quad i = 1, \dots, r. \quad (2)$$

We assume that ϵ_{ij} and ϵ_{0i} are i.i.d. $N(0, \sigma_\epsilon^2)$. The variable X_i takes a fixed value. The model parameters are α, β, X_0 and σ_ϵ^2 , and the main aim is to estimate the value X_0 . When $m_i = 1$, one has the linear calibration model [2] which is very useful in applications when there is no measurement error [2,3]. In this paper, we extend this model to the case $m_i \geq 1$ and the model defined by Equations (1)–(2) will be called the *usual calibration model with replicates* (Usual-M). It assumes that the analyte concentrations from the series of standard solutions are accurately known, so this model does not consider measurement error. Maximum likelihood estimators for Usual-M are given in Appendix 1.

Blas *et al.* [4] proposed the so-called *homoscedastic controlled calibration model* without replicates, i.e. $m_i = 1$, where X_i is a pre-fixed target value of a predictor variable (x_i), but the unobserved true value x_i can differ from the target value, that is,

$$x_i = X_i - u_i, \quad i = 1, \dots, n, \quad (3)$$

where u_i is a Berkson-type measurement error. An heteroscedastic version of Blas *et al.* [4] is studied in [5]. In chemical applications, the measurement error u_i comes from the preparation process of the standard solution. Blas *et al.* [4] incorporated the error due to the lecture of equipment and the preparation of the standard solutions and also showed that their approach is sensible to the presence of error related to the independent variable which gives better results in contrast to the Usual-M.

The measurement method to obtain Y is relatively fast, cheap, easy and inaccurate, so replicated measurement in each standard solution is feasible in the calibration model. Thus, motivated by the chemical applications with replicates on the variable Y , we propose a generalization of the controlled calibration model proposed by Blas *et al.* [4], which follows by considering the usual model defined in Equations (1)–(3) from the Berkson-type measurement error, and this new approach will be termed as the *controlled calibration model with replicates* (Proposed-M).

The assessment of robustness aspects of the parameter estimates in statistical models has been an important concern of various researchers in recent decades. The deletion methodology, which consists of studying the impact on the parameter estimates after dropping individual observations, is probably the most employed technique to detect influential observations – see Cook and Weisberg [6]. Nevertheless, research on the influence of small perturbations in the model/data on the parameter estimates has received increasing attention in recent years. This can be achieved by performing local influence analysis, a general statistical technique used to assess the stability of the estimation outputs with respect to the model inputs. Following the pioneering work of Cook [7], this area of research has received considerable attention in the recent statistical literature: see for instance, [8] and [9]. However, in many models the marginal log-likelihood function is complex and direct application of Cook's [7] approach may be very difficult, because these measures involve the first and second partial derivatives of this function. Wu and Luo [10] generalized Cook's approach and presented the influence on the maximum likelihood estimate of any regression model. Zhu and Lee [11] developed an approach to perform local influence analysis for general statistical models with missing data by working with a Q -displacement function

closely related to the conditional expectation of the complete-data log-likelihood at the E-step of the EM algorithm. Zhu *et al.* [12] developed a rigorous differential–geometrical framework on a perturbation model, named as the perturbation manifold. This approach shows that the metric tensor of the perturbation manifold provides important information about selecting an appropriate perturbation of a model and also defines new influence measures for smooth objective functions. The theory of Zhu *et al.* [12] is developed on the basis of the observed data likelihood, and it is difficult to apply to complex latent variable models that involve observed-data likelihood with intractable integrals. Recently, Chen *et al.* [13] developed the local influence method based on the method of Zhu *et al.* [12] in the context of general latent variable models.

Inspired by Zhu and Lee [11] and Lee and Xu [14], in this paper we apply Zhu and Lee’s local influence approach to our Proposed-M and it will be seen that such an approach leads to simpler influence measures.

This paper is organized in the following manner. In Section 2, the Proposed-M is presented, as well as some classical parameter estimation methods and hypotheses testing. In Section 3, we provide a brief sketch of the local influence measures developed via the approach of Zhu and Lee. Four different perturbation schemes are considered. Section 4 presents results of simulation studies to investigate the sensitivity of parameter estimates of the Proposed-M compared with the Usual-M. Section 5 presents illustrative examples with a real data set. Finally, some conclusions derived from this work is presented in Section 6.

2. The controlled calibration model with replicates

Blas *et al.* [4] defined a univariate homoscedastic controlled calibration model, where the independent variable (x_i) is unobservable, but a pre-fixed value on its surrogate is available (X_i). Their approach follows by considering controlled variables and assuming that the measurement errors have equal variances. In this work, we extended the model defined in [4] by considering replicated measurement in the response variable. Thus, the following model is defined:

$$Y_{ij} = \alpha + \beta x_i + \epsilon_{ij}, \quad j = 1, \dots, m_i \text{ and } i = 1, \dots, n, \quad (4)$$

$$x_i = X_i + u_i, \quad i = 1, \dots, n, \quad (5)$$

$$Y_{0i} = \alpha + \beta X_0 + \epsilon_{0i}, \quad i = 1, \dots, r, \quad (6)$$

or, alternatively,

$$Y_{ij} = \alpha + \beta X_i + (\epsilon_{ij} + \beta u_i), \quad j = 1, \dots, m_i \text{ and } i = 1, \dots, n, \quad (7)$$

$$Y_{0i} = \alpha + \beta X_0 + \epsilon_{0i}, \quad i = 1, \dots, r, \quad (8)$$

with the following assumptions on the random errors:

- ϵ_{ij} and ϵ_{0i} are independent $N(0, \sigma_\epsilon^2)$ random variables,
- u_i are independent $N(0, \sigma_u^2)$ random variables,
- $\text{cov}(u_i, \epsilon_{ij}) = 0$ and $\text{cov}(u_i, \epsilon_{0j}) = 0$ for all i, j .

The model parameters are $\alpha, \beta, X_0, \sigma_\epsilon^2$ and σ_u^2 and the main interest is to estimate the quantity X_0 .

Some comments are in order here. The variable X_i in Equation (7) is a pre-fixed value, and the model error ($\epsilon_{ij} + \beta u_i$) is independent of X_i (Berkson-type errors). We also define a homoscedastic controlled calibration model by considering that the errors u_i are independent and normally distributed with mean 0 and constant variance, σ_u^2 . Further, we consider replicates on both variables.

The explicit form of the maximum likelihood (ML) estimator of $\hat{X}_0$ and the EM estimator of $\hat{X}_0$ are given in Appendix 1 and Section 2.2, respectively.

An approximation for the variance of $\hat{X}_0$ is given by the Fisher information of $\theta = (\alpha, \beta, X_0, \sigma_\epsilon^2, \sigma_u^2)$ which, after some length algebraic manipulations, can be shown to be given by

$$I(\theta) = \begin{pmatrix} \frac{N}{\gamma} + \frac{r}{\sigma_\epsilon^2} & \frac{rX_0}{\sigma_\epsilon^2} + \frac{N\bar{X}}{\gamma} & \frac{r\beta}{\sigma_\epsilon^2} & 0 & 0 \\ \frac{rX_0}{\sigma_\epsilon^2} + \frac{N\bar{X}}{\gamma} & \frac{rX_0^2}{\sigma_\epsilon^2} + I_1 & \frac{k\beta X_0}{\sigma_\epsilon^2} & \frac{N\beta^3\sigma_u^2}{\gamma^2} & \frac{N\beta\sigma_u^2}{\gamma^2} \\ \frac{r\beta}{\sigma_\epsilon^2} & \frac{r\beta X_0}{\sigma_\epsilon^2} & \frac{r\beta^2}{\sigma_\epsilon^2} & 0 & 0 \\ 0 & \frac{N\beta^3\sigma_u^2}{\gamma^2} & 0 & \frac{N\beta^4}{2\gamma^2} & \frac{N\beta^2}{2\gamma^2} \\ 0 & \frac{N\beta\sigma_u^2}{\gamma^2} & 0 & \frac{N\beta^2}{2\gamma^2} & \frac{N}{2\gamma^2} + \frac{r}{2\sigma_\epsilon^4} \end{pmatrix},$$

where $I_1 = \sum_{i=1}^n m_i X_i^2 / \gamma + 2N\beta^2\sigma_u^2 / \gamma^2$, $\bar{X} = \sum_{i=1}^n m_i X_i / N$ and $N = \sum_{i=1}^n m_i$.

The variance of $\hat{X}_0$, derived from the Fisher information matrix, is given by

$$V(\hat{X}_0) = \frac{\sigma_\epsilon^2}{\beta^2} \left[\frac{1}{k} + \frac{\gamma}{N\sigma_\epsilon^2} + \frac{\gamma}{\sigma_\epsilon^2} \frac{(\bar{X} - X_0)^2}{\sum_{i=1}^n m_i (X_i - \bar{X})^2} \right], \quad (9)$$

where

$$\gamma = \beta^2\sigma_u^2 + \sigma_\epsilon^2. \quad (10)$$

We can find the variance of $\hat{X}_0$ from the Usual-M by making $\sigma_u^2 = 0$.

2.1. Inference

The approximated confidence interval for X_0 with confidence coefficient $(1 - \alpha)$, is given by

$$\left(\hat{X}_0 - z_{\alpha/2} \sqrt{\widehat{V(\hat{X}_0)}}, \hat{X}_0 + z_{\alpha/2} \sqrt{\widehat{V(\hat{X}_0)}} \right), \quad (11)$$

where $z_{\alpha/2}$ is the quantile of order $(1 - \alpha/2)$ of the standard normal distribution and $\widehat{V(\hat{X}_0)}$ is the estimator of $V(\hat{X}_0)$ from Equation (9). This confidence interval can be found using the closed form of the ML estimators from Appendix 2 or the EM estimator from Section 2.2.

Hypotheses testing: The main interest in this model is to test the hypothesis $H_0 : X_0 = X_{00}$, where X_{00} is a specified value. Note that, H_0 can be tested using the interval (11).

2.2. The EM algorithm

The maximum likelihood estimate of σ_u^2 based on the direct maximum likelihood estimator approach can supply negative values, hence, an alternative method for computing maximum likelihood estimator is called for. Due to this consideration, we demonstrate how to use the EM algorithm for maximum likelihood estimation in the Proposed-M. We note that the EM-algorithm [15] is computationally easy to implement and, moreover, guarantees no negative scale parameter estimates.

The model defined in Equations (4)–(6) can be written hierarchically as

$$Y_{ij}|x_i \stackrel{\text{ind}}{\sim} N(\alpha + \beta x_i, \sigma_\epsilon^2), \quad j = 1, \dots, m_i \text{ and } i = 1, \dots, n, \quad (12)$$

$$x_i \stackrel{\text{ind}}{\sim} N(X_i, \sigma_u^2), \quad i = 1, \dots, n, \quad (13)$$

$$Y_{0i} \stackrel{\text{ind}}{\sim} N(\alpha + \beta X_0, \sigma_\epsilon^2), \quad i = 1, \dots, r. \quad (14)$$

Thus, the complete-data log-likelihood function, which is denoted by $l_c(\boldsymbol{\theta}|\mathbf{Y}, \mathbf{Y}_0, \mathbf{x})$, with $\mathbf{Y} = (Y_1^T, \dots, Y_n^T)^T$, $\mathbf{Y}_0 = (Y_{01}, \dots, Y_{0r})^T$, $\mathbf{x} = (x_1, \dots, x_n)$ and $\mathbf{Y}_i = (Y_{i1}, \dots, Y_{im_i})$, $i = 1, \dots, n$, is given by

$$\begin{aligned} l_c(\boldsymbol{\theta}|\mathbf{Y}, \mathbf{Y}_0, \mathbf{x}) \propto & -\frac{N}{2} \log(\sigma_\epsilon^2) - \frac{N}{2} \log(\sigma_u^2) - \frac{r}{2} \log(\sigma_\epsilon^2) \\ & - \frac{1}{2} \left[\frac{1}{\sigma_\epsilon^2} \sum_{i=1}^n \sum_{j=1}^{m_i} (Y_{ij} - \alpha - \beta x_i)^2 + \frac{1}{\sigma_u^2} \sum_{i=1}^n m_i (x_i - X_i)^2 \right. \\ & \left. + \frac{1}{\sigma_\epsilon^2} \sum_{i=1}^r (Y_{0i} - \alpha - \beta X_0)^2 \right]. \end{aligned}$$

For the current value $\hat{\boldsymbol{\theta}}^{(k)}$ in the iteration k , the E-step of the EM-type algorithm requires the evaluation of $Q(\boldsymbol{\theta}|\hat{\boldsymbol{\theta}}^{(k)}) = E[l_c(\boldsymbol{\theta}|\mathbf{Y}, \mathbf{Y}_0, \mathbf{x})|\mathbf{Y}, \mathbf{Y}_0, \hat{\boldsymbol{\theta}}^{(k)}]$, where the expectation is taken with respect to the joint conditional distribution of $\mathbf{x}$ given $\mathbf{Y}$ and $\mathbf{Y}_0$. Thus, we have that

$$\begin{aligned} Q(\boldsymbol{\theta}|\hat{\boldsymbol{\theta}}^{(k)}) = & -\frac{N}{2} \log(\sigma_\epsilon^2) - \frac{N}{2} \log(\sigma_u^2) - \frac{r}{2} \log(\sigma_\epsilon^2) \\ & - \frac{1}{2} \left[\frac{1}{\sigma_\epsilon^2} \sum_{i=1}^n \sum_{j=1}^{m_i} \{ (Y_{ij} - \alpha)^2 + \beta^2 \hat{x}_i^{2(k)} - 2\beta \hat{x}_i^{(k)} (Y_{ij} - \alpha) \} \right. \\ & \left. - \frac{1}{\sigma_u^2} \sum_{i=1}^n m_i (\hat{x}_i^{2(k)} + X_i^2 - 2\hat{x}_i^{(k)} X_i) + \frac{1}{\sigma_\epsilon^2} \sum_{i=1}^r (Y_{0i} - \hat{\alpha}^{(k)} - \hat{\beta}^{(k)} X_0)^2 \right], \end{aligned}$$

where $\hat{x}_i^{(k)} = E[x_i|\mathbf{Y}, \mathbf{Y}_0, \hat{\boldsymbol{\theta}}^{(k)}]$ and $\hat{x}_i^{2(k)} = E[x_i^2|\mathbf{Y}, \mathbf{Y}_0, \hat{\boldsymbol{\theta}}^{(k)}]$, for $i = 1, \dots, n$, are given by

$$\hat{x}_i^{(k)} = X_i + \frac{\hat{\beta}^{(k)} \hat{\sigma}_u^{2(k)}}{\hat{\gamma}_i^{(k)}} \sum_{j=1}^{m_i} (Y_{ij} - \hat{\alpha}^{(k)} - \hat{\beta}^{(k)} X_i), \quad (15)$$

$$\begin{aligned} \hat{x}_i^{2(k)} = & \frac{\hat{\sigma}_\epsilon^{2(k)} \hat{\sigma}_u^{2(k)}}{\hat{\gamma}_i^{(k)}} + X_i^2 + \frac{\hat{\beta}^{2(k)} \hat{\sigma}_u^{4(k)}}{\hat{\gamma}_i^{2(k)}} \left(\sum_{j=1}^{m_i} (Y_{ij} - \hat{\alpha}^{(k)} - \hat{\beta}^{(k)} X_i) \right)^2 \\ & + \frac{2\hat{\beta}^{(k)} \hat{\sigma}_u^{2(k)} X_i}{\hat{\gamma}_i^{(k)}} \sum_{j=1}^{m_i} (Y_{ij} - \hat{\alpha}^{(k)} - \hat{\beta}^{(k)} X_i), \end{aligned} \quad (16)$$

with $\hat{\gamma}_i^{(k)} = \hat{\sigma}_\epsilon^{2(k)} + m_i \hat{\beta}^{2(k)} \hat{\sigma}_u^{2(k)}$.

The M steps then conditionally maximize $Q(\theta|\hat{\theta}^{(k)})$ with respect to $\theta^{(k)}$, leading to a new estimate $\theta^{(k+1)}$, as described below:

$$\hat{X}_0^{(k+1)} = \frac{\bar{Y}_0 - \hat{\alpha}^{(k)}}{\hat{\beta}^{(k)}}, \quad (17)$$

$$\hat{\beta}^{(k+1)} = \frac{\sum_{i=1}^n \sum_{j=1}^m \hat{x}_i^{(k)} (Y_{ij} - \hat{\alpha}^{(k)})}{\sum_{i=1}^n m_i \hat{x}_i^{(k)}}, \quad (18)$$

$$\hat{\alpha}^{(k+1)} = \frac{\sum_{i=1}^n \sum_{j=1}^m Y_{ij} - \hat{\beta}^{(k+1)} \sum_{i=1}^n m_i \hat{x}_i^{(k)}}{\sum_{i=1}^n m_i}, \quad (19)$$

$$\hat{\sigma}_u^{2(k+1)} = \frac{1}{N} \sum_{i=1}^n m_i (\hat{x}_i^{(k)} + X_i^2 - 2\hat{x}_i^{(k)} X_i), \quad (20)$$

$$\begin{aligned} \hat{\sigma}_\epsilon^{2(k+1)} = \frac{1}{r+N} & \left[\sum_{i=1}^n \sum_{j=1}^{m_i} ((Y_{ij} - \hat{\alpha}^{(k+1)})^2 + \hat{\beta}^{2(k+1)} \hat{x}_i^{(k)} \right. \\ & \left. - 2\hat{\beta}^{(k+1)} \hat{x}_i^{(k)} (Y_{ij} - \hat{\alpha}^{(k+1)})) + \sum_{i=1}^r (Y_{0i} - \hat{\alpha}^{(k)} - \hat{\beta}^{(k)} X_0)^2 \right]. \end{aligned} \quad (21)$$

The above algorithm is iterated until a suitable convergence rule is satisfied, e.g. $\|\theta^{(k+1)} - \theta^{(k)}\|$ is sufficiently small. Note that this algorithm is not computationally expensive and guarantees no negative scale parameter estimates for σ_u^2 and σ_ϵ^2 , once substituting $\hat{x}_i^{(k)}$ and $\hat{x}_i^{2(k)}$ into Equations (19) and (20), we have that

$$\begin{aligned} \hat{\sigma}_u^{2(k+1)} &= \frac{1}{N} \sum_{i=1}^n m_i \left\{ \frac{\sigma_\epsilon^{2(k)} \sigma_u^{2(k)}}{\gamma_i^{(k)}} + \frac{\beta^{2(k+1)} \sigma_u^{2(k)}}{\gamma_i^{(k)}} \left(\sum_{j=1}^{m_i} (Y_{ij} - \alpha^{(k+1)} - \beta^{(k+1)} X_i) \right)^2 \right\}, \\ \hat{\sigma}_\epsilon^{2(k+1)} &= \frac{1}{r+N} \left[\sum_{i=1}^n \sum_{j=1}^{m_i} \left\{ \frac{\beta^{2(k+1)} \sigma_u^{2(k)} \sigma_\epsilon^{2(k)}}{\gamma_i^{(k)}} + (\beta^{2(k+1)} \hat{x}_i^{(k)} - Y_{ij} + \hat{\alpha}^{(k+1)})^2 \right\} \right. \\ & \quad \left. + \left(\sum_{j=1}^{m_i} (Y_{ij} - \alpha^{(k+1)} - \beta^{(k+1)} X_i) \right)^2 \right]. \end{aligned}$$

The iterative algorithm can begin with the estimates from the explicit form of the ML estimators given in Appendix 2.

Influence diagnostic techniques consist in evaluating the sensitivity of the parameter estimates of a particular model when perturbation occurs in the data set or in the assumptions of the model. Case deletion [16] is the simplest way of perturbing the model or the data. This method is also known as the global influence method and it consists in analyzing one or more fitted models after excluding observations. In the next section, we will discuss the local influence method based on the conditional expectation of the complete-data log-likelihood function $Q(\theta|\hat{\theta}^{(k)})$ proposed by Zhu and Lee [11], which analyzes the sensitivity of the parameter estimates without deleting the observations.

In chemical applications, the small number of standard solutions makes the application of some statistical analysis difficult, for instance, the diagnostic methods, but considering replications this

constraint is overtaken. The Proposed-M considers replicated data which makes the local influence approach a feasible technique.

3. The local influence approach

Consider a perturbation vector $\omega = (w_1, \dots, w_g)^T$ which is in some open set $\Omega \subseteq \mathbb{R}^g$. Let $\ell_c(\theta, \omega | \mathbf{Y}, \mathbf{Y}_0, \mathbf{x})$, $\theta \in \mathbb{R}^h$, be the complete-data log-likelihood of the perturbed model. We assume that there is a ω_0 such that $\ell_c(\theta, \omega_0 | \mathbf{Y}, \mathbf{Y}_0, \mathbf{x}) = \ell_c(\theta | \mathbf{Y}, \mathbf{Y}_0, \mathbf{x})$ for all θ . Let $\hat{\theta}(\omega)$ be the maximum likelihood estimate of Q for the perturbed model which maximizes the function

$$Q(\theta, \omega | \hat{\theta}) = E[\ell_c(\theta, \omega | \mathbf{Y}, \mathbf{Y}_0, \mathbf{x}) | \mathbf{Y}, \mathbf{Y}_0, \hat{\theta}].$$

Obviously, $\hat{\theta}(\omega_0) = \hat{\theta}$. The influence graph is defined as $\alpha(\omega) = (\omega^T, f_Q(\omega))^T$, where $f_Q(\omega)$ is the Q -displacement function defined as

$$f_Q(\omega) = 2[Q(\hat{\theta} | \hat{\theta}) - Q(\hat{\theta}(\omega) | \hat{\theta})].$$

Following the approach developed in [7,11], the normal curvature $C_{f_Q, \mathbf{d}}$ of $\alpha(\omega)$ at ω_0 in the direction of some unit vector $\mathbf{d}$ can be used to summarize the local behaviour of the Q -displacement function. It can be shown (see [11]) that

$$C_{f_Q, \mathbf{d}} = -2\mathbf{d}^T \ddot{Q}_{\omega_0} \mathbf{d} \quad \text{and} \quad -\ddot{Q}_{\omega_0} = \Delta_{\omega_0}^T \{-\ddot{Q}_{\theta}(\hat{\theta})\}^{-1} \Delta_{\omega_0},$$

where

$$\ddot{Q}_{\theta}(\hat{\theta}) = \frac{\partial^2 Q(\theta | \hat{\theta})}{\partial \theta \partial \theta^T} \Big|_{\theta=\hat{\theta}} \quad \text{and} \quad \Delta_{\omega} = \frac{\partial^2 Q(\theta, \omega | \hat{\theta})}{\partial \theta \partial \omega^T} \Big|_{\theta=\hat{\theta}(\omega)}.$$

Under mild regularity conditions [11] $-\ddot{Q}_{\omega_0}$ is semi-positive definite. Thus, from the spectral decomposition of a symmetric matrix,

$$-2\ddot{Q}_{\omega_0} = \sum_{k=1}^g \lambda_k \mathbf{e}_k \mathbf{e}_k',$$

where $(\lambda_1, \mathbf{e}_1), \dots, (\lambda_g, \mathbf{e}_g)$ are the eigenvalue–eigenvector pairs for the matrix $-2\ddot{Q}_{\omega_0}$ with $\lambda_1 \geq \dots \geq \lambda_h, \lambda_{h+1} = \dots = \lambda_g = 0$ and $\mathbf{e}_1, \dots, \mathbf{e}_g$ are the elements of the associated orthonormal basis. On the basis of results given in [7], $C_{f_Q, \mathbf{d}} = \lambda_1$, is the largest eigenvalue. Thus, the eigenvector $\mathbf{e}_1 = \mathbf{d}_{\max}$ corresponding to λ_1 gives important information for assessing local influence of a minor perturbation. When g is larger, it is computationally expensive to obtain the eigenvalues–eigenvectors $\{(\lambda_k, \mathbf{e}_k), k = 1, \dots, g\}$. Moreover, as λ_1 can take any positive value, sometimes it is not easy to judge its size. In order to reduce the computational burden and to aid interpretation, Zhu and Lee [11] consider an aggregated contribution vector of all those eigenvectors with nonzero eigenvalues. Introducing some further notation, let $\tilde{\lambda}_k = \lambda_k / (\lambda_1 + \dots + \lambda_h)$, $\mathbf{e}_k^2 = (e_{k1}^2, \dots, e_{kg}^2)$ and

$$M(0) = \sum_{k=1}^h \tilde{\lambda}_k \mathbf{e}_k^2,$$

with the l th component of $M(0)$, denoted by $M(0)_l$, equal to $\sum_{k=1}^h \tilde{\lambda}_k e_{kl}^2$. Assessment of influential cases is based on the visual inspection of the $\{M(0)_l, l = 1, \dots, g\}$ plotted against the index l . Inspired by the work of Poon and Poon [17] in using a conformal normal curvature to modify

the normal curvature in [7], one can obtain $M(0)_l$ via $B_{f_Q, \mathbf{u}_l} = -2\mathbf{u}_l^T \ddot{Q}_{\omega_0} \mathbf{u}_l / \text{tr}[-2\ddot{Q}_{\omega_0}]$, where $\mathbf{u}_l$ is a column vector in $\mathbb{R}^g$ with the l th entry equal to one and all other entries zero. Note that the evaluation of $B_{f_Q, \mathbf{u}_l}$ is very simple, and it is not necessary to compute the eigenvalues–eigenvectors of $\ddot{Q}_{\omega_0}$. We refer the reader to Poon and Poon [11] for other theoretical properties of $B_{f_Q, \mathbf{u}_l}$, such as its invariance under reparametrizations of θ .

Let $\bar{M}(0)$ and $SM(0)$ be the mean and standard error of $\{M(0)_l, l = 1, \dots, g\}$. Clearly, $\bar{M}(0) = 1/g$. It is natural to regard observations with $M(0)_l$ -values that are significantly larger than $\bar{M}(0)$ as influential. This motivates the use of $\bar{M}(0)$ in defining a benchmark. However, one clearly can use different functions of $\bar{M}(0)$. For example, in the context of the approach of Cook [7], the use of $2\bar{M}(0)$ is proposed in [17], which is similar to the one often employed for identifying leverage in regression analysis. In order to take into account the variation in $M(0)_l$, the use of $\bar{M}(0) + 2SM(0)$ is proposed in [11], and the use of $\bar{M}(0) + c^*SM(0)$ is proposed in [14], where c^* is a constant taken to be 2 or higher. This strategy worked well in the empirical studies that we have conducted.

3.1. The Hessian matrix $\ddot{Q}_\theta(\hat{\theta})$

In order to obtain the diagnostic measures for local influence of a particular perturbation scheme, it is necessary to compute the symmetric matrix $\ddot{Q}_\theta(\hat{\theta}) = \partial^2 Q(\theta|\hat{\theta}) / \partial \theta \partial \theta^T$, where $\theta = (\alpha, \beta, X_0, \sigma_\epsilon^2, \sigma_u^2)$. Hence, we have that

$$\ddot{Q}_\theta(\hat{\theta}) = \begin{pmatrix} \ddot{Q}_{11} & \ddot{Q}_{12} & \ddot{Q}_{13} & \ddot{Q}_{14} & 0 \\ & \ddot{Q}_{22} & \ddot{Q}_{23} & \ddot{Q}_{24} & 0 \\ & & \ddot{Q}_{33} & \ddot{Q}_{34} & 0 \\ & & & \ddot{Q}_{44} & 0 \\ \text{symmetric} & & & & \ddot{Q}_{55} \end{pmatrix},$$

where

$$\begin{aligned} \ddot{Q}_{11} &= \frac{\partial^2 Q(\theta|\hat{\theta})}{\partial \alpha \partial \alpha} = -\frac{(N+r)}{\sigma_\epsilon^2}, & \ddot{Q}_{12} &= \frac{\partial^2 Q(\theta|\hat{\theta})}{\partial \alpha \partial \beta} = -\frac{(rX_0 + \sum_{i=1}^n m_i \hat{x}_i^{(k)})}{\sigma_\epsilon^2}, \\ \ddot{Q}_{13} &= \frac{\partial^2 Q(\theta|\hat{\theta})}{\partial \alpha \partial X_0} = -\frac{r\beta}{\sigma_\epsilon^2}, & \ddot{Q}_{22} &= \frac{\partial^2 Q(\theta|\hat{\theta})}{\partial \beta \partial \beta} = -\frac{(rX_0^2 + \sum_{i=1}^n m_i \hat{x}_i^{(k)2})}{\sigma_\epsilon^2}, \\ \ddot{Q}_{14} &= \frac{\partial^2 Q(\theta|\hat{\theta})}{\partial \alpha \partial \sigma_\epsilon^2} = \frac{1}{\sigma_\epsilon^4} \left\{ N\alpha - \sum_{i=1}^n \sum_{j=1}^{m_i} (y_{ij} - \beta \hat{x}_i^{(k)}) + r(\alpha + \beta X_0 - \bar{Y}_0) \right\}, \\ \ddot{Q}_{23} &= \frac{\partial^2 Q(\theta|\hat{\theta})}{\partial \beta \partial X_0} = -\frac{r}{\sigma_\epsilon^2} (\alpha + 2\beta X_0 - \bar{Y}_0), \\ \ddot{Q}_{24} &= \frac{\partial^2 Q(\theta|\hat{\theta})}{\partial \beta \partial \sigma_\epsilon^2} = \frac{1}{\sigma_\epsilon^4} \left\{ \sum_{i=1}^n \sum_{j=1}^{m_i} (\beta \hat{x}_i^{(k)2} - \hat{x}_i^{(k)} (Y_{ij} - a)) + rX_0(\alpha + \beta X_0 - \bar{Y}_0) \right\}, \\ \ddot{Q}_{33} &= \frac{\partial^2 Q(\theta|\hat{\theta})}{\partial X_0 \partial X_0} = -\frac{r\beta^2}{\sigma_\epsilon^2}, & \ddot{Q}_{34} &= \frac{\partial^2 Q(\theta|\hat{\theta})}{\partial X_0 \partial \sigma_\epsilon^2} = \frac{r\beta}{\sigma_\epsilon^4} (\alpha + \beta X_0 - \bar{Y}_0), \end{aligned}$$

$$\begin{aligned}
\ddot{Q}_{44} &= \frac{\partial^2 Q(\boldsymbol{\theta}|\hat{\boldsymbol{\theta}})}{\partial \sigma_\epsilon^2 \partial \sigma_u^2} = -\frac{1}{2\sigma_\epsilon^6 \sigma_u^2} (-\sigma_u^2 \sigma_\epsilon^2 (N+r) + 2N\alpha^2 \sigma_u^2 \\
&\quad - 2 \sum_{i=1}^n \sum_{j=1}^{m_i} [-\sigma_u^2 (Y_{ij}^2 - 2\alpha Y_{ij} + \beta^2 \hat{x}_i^{(k)} - 2\beta x_i Y_{ij} + 2\beta \alpha \hat{x}_i^{(k)})] \\
&\quad + 2r\sigma_u^2 (\alpha^2 + 2\alpha\beta X_0 + \beta^2 X_0^2) + 2\sigma_u^2 \sum_{i=1}^r Y_{0i} (Y_{0i} - 2\alpha - 2\beta X_0)), \\
\ddot{Q}_{55} &= \frac{\partial^2 Q(\boldsymbol{\theta}|\hat{\boldsymbol{\theta}})}{\partial \sigma_u^2 \partial \sigma_u^2} = \frac{N}{2\sigma_u^4} - \frac{1}{\sigma_\epsilon^6} \sum_{i=1}^n m_i (X_i^2 + \hat{x}_i^{(k)} - 2\hat{x}_i^{(k)} X_i).
\end{aligned}$$

3.2. Interesting perturbation schemes

In this section, we consider four different perturbation schemes for the model defined in Equations (12)–(14) (the Proposed-M).

3.2.1. Case weight perturbation

First, we consider an arbitrary attribution of weights for the expected value of the complete-data log-likelihood function (perturbed Q-function), which may capture observations with outstanding contribution on the log-likelihood function and that may exercise high influence on the maximum likelihood estimates. This is represented by writing

$$\begin{aligned}
Q(\boldsymbol{\theta}, \boldsymbol{\omega}, \boldsymbol{\omega}'|\hat{\boldsymbol{\theta}}) &= E[\ell_c(\boldsymbol{\theta}, \boldsymbol{\omega}, \boldsymbol{\omega}'|\mathbf{Y}, \mathbf{Y}_0, \mathbf{x})] \\
&= \sum_{i=1}^n \sum_{j=1}^{m_i} \omega_{ij} E[\ell_{ij}(\boldsymbol{\theta}|\mathbf{Y}, \mathbf{Y}_0, \mathbf{x})] + \sum_{i=1}^r \omega'_i E[\ell_{0i}(\boldsymbol{\theta}|\mathbf{Y}, \mathbf{Y}_0, \mathbf{x})] \\
&= \sum_{i=1}^n \sum_{j=1}^{m_i} Q_{1ij}(\boldsymbol{\theta}, \boldsymbol{\omega}|\hat{\boldsymbol{\theta}}) + \sum_{i=1}^r Q_{2i}(\boldsymbol{\theta}, \boldsymbol{\omega}'|\hat{\boldsymbol{\theta}}),
\end{aligned}$$

where $\ell_c(\boldsymbol{\theta}, \boldsymbol{\omega}, \boldsymbol{\omega}'|\mathbf{Y}, \mathbf{Y}_0, \mathbf{x}) = \sum_{i=1}^n \sum_{j=1}^{m_i} \omega_{ij} \ell_{ij}(\boldsymbol{\theta}|\mathbf{Y}, \mathbf{Y}_0, \mathbf{x}) + \sum_{i=1}^r \omega'_i \ell_{0i}(\boldsymbol{\theta}|\mathbf{Y}, \mathbf{Y}_0, \mathbf{x})$, which is the sum of the first and the second stages log-likelihood functions, respectively.

We have that $\boldsymbol{\omega} = (\omega_{11}, \dots, \omega_{1m_1}, \omega_{21}, \dots, \omega_{2m_2}, \dots, \omega_{n1}, \dots, \omega_{1m_n})^T$ is an $(\sum_{i=1}^n m_i) \times 1$ vector and $\boldsymbol{\omega}' = (\omega'_1, \dots, \omega'_r)$ is an $r \times 1$ vector. The perturbed Q -function related to the first and second stages are $Q_{1ij}(\boldsymbol{\theta}, \boldsymbol{\omega}|\hat{\boldsymbol{\theta}}) = \omega_{ij} Q_{1ij}(\boldsymbol{\theta}|\hat{\boldsymbol{\theta}})$ and $Q_{2i}(\boldsymbol{\theta}, \boldsymbol{\omega}'|\hat{\boldsymbol{\theta}}) = \omega'_i Q_{2i}(\boldsymbol{\theta}|\hat{\boldsymbol{\theta}})$, respectively, with $Q_{1ij}(\boldsymbol{\theta}|\hat{\boldsymbol{\theta}})$ and $Q_{2i}(\boldsymbol{\theta}|\hat{\boldsymbol{\theta}})$ given by

$$\begin{aligned}
Q_{1ij}(\boldsymbol{\theta}|\hat{\boldsymbol{\theta}}) &= -\frac{1}{2} \log(\sigma_\epsilon^2) - \frac{1}{2} \log(\sigma_u^2) - \frac{1}{2\sigma_\epsilon^2} \{(Y_{ij} - \alpha)^2 + \beta^2 \hat{x}_i^2 - 2\beta \hat{x}_i (Y_{ij} - \alpha)\} \\
&\quad - \frac{1}{\sigma_u^2} (x_i^2 + X_i^2 - 2\hat{x}_i X_i), \\
Q_{2i}(\boldsymbol{\theta}|\hat{\boldsymbol{\theta}}) &= -\frac{1}{2} \log(\sigma_\epsilon^2) - \frac{1}{2\sigma_\epsilon^2} (Y_{0i} - \alpha - \beta X_0)^2.
\end{aligned}$$

Under this perturbation scheme, the non-perturbed model is obtained when $\boldsymbol{\omega}_0 = (1, \dots, 1)^T$, in this case the matrix $\boldsymbol{\Delta}_{\omega_0} = (\partial^2 Q(\boldsymbol{\theta}, (\boldsymbol{\omega}, \boldsymbol{\omega}')|\hat{\boldsymbol{\theta}}) / \partial \boldsymbol{\theta} \partial (\boldsymbol{\omega}, \boldsymbol{\omega}')^T) |_{(\boldsymbol{\omega}, \boldsymbol{\omega}') = \boldsymbol{\omega}_0}$. The first $\sum_{i=1}^n m_i$ column

vectors have the following elements:

$$\begin{aligned} \left. \frac{\partial Q_{1ij}(\boldsymbol{\theta}, \boldsymbol{\omega} | \hat{\boldsymbol{\theta}})}{\partial \alpha \partial \omega_{ij}} \right|_{\omega_0} &= \frac{1}{\sigma_\epsilon^2} (Y_{ij} - \beta \hat{x}_i - \alpha), \\ \left. \frac{\partial Q_{1ij}(\boldsymbol{\theta}, \boldsymbol{\omega} | \hat{\boldsymbol{\theta}})}{\partial \beta \partial \omega_{ij}} \right|_{\omega_0} &= -\frac{1}{\sigma_\epsilon^2} (\beta \hat{x}_i^2 - \hat{x}_i (Y_{ij} - \alpha)), \\ \left. \frac{\partial Q_{1ij}(\boldsymbol{\theta}, \boldsymbol{\omega} | \hat{\boldsymbol{\theta}})}{\partial \sigma_\epsilon^2 \partial \omega_{ij}} \right|_{\omega_0} &= -\frac{1}{2\sigma_\epsilon^2} + \frac{1}{2\sigma_\epsilon^4} (Y_{ij}^2 - 2Y_{ij}\alpha + \beta^2 \hat{x}_i^2 - 2\beta \hat{x}_i (Y_{ij} - \alpha) + \alpha^2), \\ \left. \frac{\partial Q_{1ij}(\boldsymbol{\theta}, \boldsymbol{\omega} | \hat{\boldsymbol{\theta}})}{\partial \sigma_u^2 \partial \omega_{ij}} \right|_{\omega_0} &= -\frac{1}{2\sigma_\epsilon^2} + \frac{1}{2\sigma_\epsilon^4} (\hat{x}_i^2 - 2\hat{x}_i X_i + X_i^2), \quad \left. \frac{\partial Q_{1ij}(\boldsymbol{\theta}, \boldsymbol{\omega} | \hat{\boldsymbol{\theta}})}{\partial X_0 \partial \omega_{ij}} \right|_{\omega_0} = 0, \end{aligned}$$

for $j = 1, \dots, m_i$, $i = 1, \dots, n$, where $\hat{x}_i$ and $\hat{x}_i^2$ were defined in Equations (15) and (16), respectively. The last r column vectors of $\boldsymbol{\Delta}_{\omega_0}$ have the following elements:

$$\begin{aligned} \left. \frac{\partial Q_{2i}(\boldsymbol{\theta}, \boldsymbol{\omega}' | \hat{\boldsymbol{\theta}})}{\partial \alpha \partial \omega'_i} \right|_{\omega_0} &= \frac{1}{\sigma_\epsilon^2} (Y_{0i} - \beta X_0 - \alpha), \\ \left. \frac{\partial Q_{2i}(\boldsymbol{\theta}, \boldsymbol{\omega}' | \hat{\boldsymbol{\theta}})}{\partial \beta \partial \omega'_i} \right|_{\omega_0} &= \frac{1}{\sigma_\epsilon^2} (Y_{0i} - \beta X_0 - \alpha) X_0, \\ \left. \frac{\partial Q_{2i}(\boldsymbol{\theta}, \boldsymbol{\omega}' | \hat{\boldsymbol{\theta}})}{\partial X_0 \partial \omega'_i} \right|_{\omega_0} &= \frac{1}{\sigma_\epsilon^2} (Y_{0i} - \beta X_0 - \alpha) \beta, \\ \left. \frac{\partial Q_{2i}(\boldsymbol{\theta}, \boldsymbol{\omega}' | \hat{\boldsymbol{\theta}})}{\partial \sigma_\epsilon^2 \partial \omega'_i} \right|_{\omega_0} &= -\frac{1}{2\sigma_\epsilon^2} + \frac{1}{2\sigma_\epsilon^4} (Y_{0i} - \beta X_0 - \alpha)^2, \\ \left. \frac{\partial Q_{2i}(\boldsymbol{\theta}, \boldsymbol{\omega}' | \hat{\boldsymbol{\theta}})}{\partial \sigma_u^2 \partial \omega'_i} \right|_{\omega_0} &= 0, \end{aligned}$$

for $i = 1, \dots, r$.

Response variable perturbation We investigate the sensitivity of the controlled calibration model in response variable by perturbing all the observations simultaneously. Under such a perturbation scheme, the original response variables $\mathbf{Y} = (\mathbf{Y}_1^T, \dots, \mathbf{Y}_n^T)^T$, where $\mathbf{Y}_i = (Y_{i1}, \dots, Y_{im_i})$, $\mathbf{Y}_0 = (Y_{01}, \dots, Y_{0r})$ are introduced by replacing Y_{ij} by $Y_{ij\omega} = Y_{ij} + \omega_{ij}$, and Y_{0i} by $Y_{0i\omega'} = Y_{0i} + \omega'_i$. In this case, $\boldsymbol{\omega}_0 = \mathbf{0}$ and

$$\begin{aligned} Q_{1ij}(\boldsymbol{\theta}, \boldsymbol{\omega} | \hat{\boldsymbol{\theta}}) &= -\frac{1}{2} \log(\sigma_\epsilon^2) - \frac{1}{2} \log(\sigma_u^2) - \frac{1}{\sigma_u^2} (\hat{x}_i^2 + X_i^2 - 2\hat{x}_i X_i) \\ &\quad - \frac{1}{2\sigma_\epsilon^2} \{ (Y_{ij\omega} - \alpha)^2 + \beta^2 \hat{x}_i^2 - 2\beta \hat{x}_i (Y_{ij\omega} - \alpha) \}, \\ Q_{2i}(\boldsymbol{\theta}, \boldsymbol{\omega}' | \hat{\boldsymbol{\theta}}) &= -\frac{1}{2} \log(\sigma_\epsilon^2) - \frac{1}{2\sigma_\epsilon^2} (Y_{0i\omega'} - \alpha - \beta X_0)^2. \end{aligned}$$

The matrix Δ_{ω_0} has the first $\sum_{i=1}^n m_i$ columns with elements given by

$$\begin{aligned} \left. \frac{\partial Q_{1ij}(\boldsymbol{\theta}, \boldsymbol{\omega}|\hat{\boldsymbol{\theta}})}{\partial \alpha \partial \omega_{ij}} \right|_{\omega_0} &= \frac{1}{\sigma_\epsilon^2}, \quad \left. \frac{\partial Q_{1ij}(\boldsymbol{\theta}, \boldsymbol{\omega}|\hat{\boldsymbol{\theta}})}{\partial \beta \partial \omega_{ij}} \right|_{\omega_0} = \frac{\hat{x}_i}{\sigma_\epsilon^2}, \quad \left. \frac{\partial Q_{1ij}(\boldsymbol{\theta}, \boldsymbol{\omega}|\hat{\boldsymbol{\theta}})}{\partial X_0 \partial \omega_{ij}} \right|_{\omega_0} = 0, \\ \left. \frac{\partial Q_{1ij}(\boldsymbol{\theta}, \boldsymbol{\omega}|\hat{\boldsymbol{\theta}})}{\partial \sigma_\epsilon^2 \partial \omega_{ij}} \right|_{\omega_0} &= \frac{1}{\sigma_\epsilon^4} (Y_{ij} - \beta \hat{x}_i - \alpha), \quad \left. \frac{\partial Q_{1ij}(\boldsymbol{\theta}, \boldsymbol{\omega}|\hat{\boldsymbol{\theta}})}{\partial \sigma_u^2 \partial \omega_{ij}} \right|_{\omega_0} = 0, \end{aligned}$$

for $j = 1, \dots, m_i$, $i = 1, \dots, n$. The last r columns of Δ_{ω_0} have the following elements:

$$\begin{aligned} \left. \frac{\partial Q_{2i}(\boldsymbol{\theta}, \boldsymbol{\omega}'|\hat{\boldsymbol{\theta}})}{\partial \alpha \partial \omega'_i} \right|_{\omega_0} &= \frac{1}{\sigma_\epsilon^2}, \quad \left. \frac{\partial Q_{2i}(\boldsymbol{\theta}, \boldsymbol{\omega}'|\hat{\boldsymbol{\theta}})}{\partial \beta \partial \omega'_i} \right|_{\omega_0} = \frac{X_0}{\sigma_\epsilon^2}, \quad \left. \frac{\partial Q_{2i}(\boldsymbol{\theta}, \boldsymbol{\omega}'|\hat{\boldsymbol{\theta}})}{\partial X_0 \partial \omega'_i} \right|_{\omega_0} = \frac{\beta}{\sigma_\epsilon^2}, \\ \left. \frac{\partial Q_{2i}(\boldsymbol{\theta}, \boldsymbol{\omega}'|\hat{\boldsymbol{\theta}})}{\partial \sigma_\epsilon^2 \partial \omega'_i} \right|_{\omega_0} &= \frac{1}{\sigma_\epsilon^4} (Y_{0i} - \beta X_0 - \alpha), \quad \left. \frac{\partial Q_{2i}(\boldsymbol{\theta}, \boldsymbol{\omega}'|\hat{\boldsymbol{\theta}})}{\partial \sigma_u^2 \partial \omega'_i} \right|_{\omega_0} = 0 \end{aligned}$$

for $i = 1, \dots, r$.

3.2.2. Perturbation of the variance σ_ϵ^2

To study the effects of departures from the assumption regarding the variance σ_ϵ^2 , we consider the perturbation $\sigma_{\epsilon\omega_i}^2 = \omega_i^{-1} \sigma_\epsilon^2$, which corresponds to the distribution of the random error ϵ being heteroscedastic. Under this perturbation scheme, the non-perturbed model is obtained when $\boldsymbol{\omega}_0 = (1, \dots, 1)^T$. In this case, we have that

$$\begin{aligned} Q_{1ij}(\boldsymbol{\theta}, \boldsymbol{\omega}|\hat{\boldsymbol{\theta}}) &= -\frac{1}{2} \log(\sigma_{\epsilon\omega_i}^2) - \frac{1}{2} \log(\sigma_u^2) - \frac{1}{\sigma_u^2} (\hat{x}_i^2 + X_i^2 - 2\hat{x}_i X_i) \\ &\quad - \frac{1}{2\sigma_{\epsilon\omega_i}^2} \{ (Y_{ij} - \alpha)^2 + \beta^2 \hat{x}_i^2 - 2\beta \hat{x}_i (Y_{ij} - \alpha) \}, \\ Q_{2i}(\boldsymbol{\theta}, \boldsymbol{\omega}'|\hat{\boldsymbol{\theta}}) &= -\frac{1}{2} \log(\sigma_{\epsilon\omega_i}^2) - \frac{1}{2\sigma_{\epsilon\omega_i}^2} (Y_{0i} - \alpha - \beta X_0)^2. \end{aligned}$$

The matrix Δ_{ω_0} has the first $\sum_{i=1}^n m_i$ columns with elements given by

$$\begin{aligned} \left. \frac{\partial Q_{1ij}(\boldsymbol{\theta}, \boldsymbol{\omega}|\hat{\boldsymbol{\theta}})}{\partial \alpha \partial \omega_{ij}} \right|_{\omega_0} &= -\frac{\alpha}{\sigma_\epsilon^2} + \frac{1}{\sigma_\epsilon^2} (Y_{ij} - \beta \hat{x}_i), \\ \left. \frac{\partial Q_{1ij}(\boldsymbol{\theta}, \boldsymbol{\omega}|\hat{\boldsymbol{\theta}})}{\partial \beta \partial \omega_{ij}} \right|_{\omega_0} &= -\frac{1}{\sigma_\epsilon^2} (\beta \hat{x}_i^2 - \hat{x}_i Y_{ij} + \alpha \hat{x}_i), \quad \left. \frac{\partial Q_{1ij}(\boldsymbol{\theta}, \boldsymbol{\omega}|\hat{\boldsymbol{\theta}})}{\partial X_0 \partial \omega_{ij}} \right|_{\omega_0} = 0, \\ \left. \frac{\partial Q_{1ij}(\boldsymbol{\theta}, \boldsymbol{\omega}|\hat{\boldsymbol{\theta}})}{\partial \sigma_\epsilon^2 \partial \omega_{ij}} \right|_{\omega_0} &= \frac{\alpha^2}{2\sigma_\epsilon^2} + \frac{1}{2\sigma_\epsilon^4} (Y_{ij}^2 - 2\alpha Y_{ij} + \beta^2 \hat{x}_i^2 - \beta \hat{x}_i Y_{ij} + 2\alpha \beta \hat{x}_i), \\ \left. \frac{\partial Q_{1ij}(\boldsymbol{\theta}, \boldsymbol{\omega}|\hat{\boldsymbol{\theta}})}{\partial \sigma_u^2 \partial \omega_{ij}} \right|_{\omega_0} &= 0 \end{aligned}$$

for $j = 1, \dots, m_i, i = 1, \dots, n$. The last r columns of $\mathbf{\Delta}_{\omega_0}$ have the following elements:

$$\begin{aligned} \left. \frac{\partial Q_{2i}(\boldsymbol{\theta}, \boldsymbol{\omega}' | \hat{\boldsymbol{\theta}})}{\partial \alpha \partial \omega'_i} \right|_{\omega'_0} &= \frac{1}{\sigma_\epsilon^2} (Y_{0i} - \alpha - \beta X_0), \\ \left. \frac{\partial Q_{2i}(\boldsymbol{\theta}, \boldsymbol{\omega}' | \hat{\boldsymbol{\theta}})}{\partial \beta \partial \omega'_i} \right|_{\omega'_0} &= \frac{1}{\sigma_\epsilon^2} (Y_{0i} - \alpha - \beta X_0) X_0, \\ \left. \frac{\partial Q_{2i}(\boldsymbol{\theta}, \boldsymbol{\omega}' | \hat{\boldsymbol{\theta}})}{\partial X_0 \partial \omega'_i} \right|_{\omega'_0} &= \frac{1}{\sigma_\epsilon^2} (Y_{0i} - \alpha - \beta X_0) \beta, \\ \left. \frac{\partial Q_{2i}(\boldsymbol{\theta}, \boldsymbol{\omega}' | \hat{\boldsymbol{\theta}})}{\partial \sigma_\epsilon^2 \partial \omega'_i} \right|_{\omega'_0} &= \frac{1}{2\sigma_\epsilon^2} (Y_{0i} - \alpha - \beta X_0)^2, \\ \left. \frac{\partial Q_{2i}(\boldsymbol{\theta}, \boldsymbol{\omega}' | \hat{\boldsymbol{\theta}})}{\partial \sigma_u^2 \partial \omega'_i} \right|_{\omega'_0} &= 0. \end{aligned}$$

for $i = 1, \dots, r$.

3.2.3. Perturbation of the variance σ_u^2

To study the effects of departures from the assumption regarding the variance σ_u^2 , we consider the perturbation $\sigma_{u\omega_i}^2 = \omega_i^{-1} \sigma_u^2$, which corresponds to the distribution of the random error u being heteroscedastic. In this case $\boldsymbol{\omega}_0 = (1, \dots, 1)^T$,

$$\begin{aligned} Q_{1ij}(\boldsymbol{\theta}, \boldsymbol{\omega} | \hat{\boldsymbol{\theta}}) &= -\frac{1}{2} \log(\sigma_\epsilon^2) - \frac{1}{2} \log(\sigma_{u\omega_i}^2) - \frac{1}{\sigma_{u\omega_i}^2} (\hat{x}_i^2 + X_i^2 - 2\hat{x}_i X_i) \\ &\quad - \frac{1}{2\sigma_\epsilon^2} \{ (Y_{ij} - \alpha)^2 + \beta^2 \hat{x}_i^2 - 2\beta \hat{x}_i (Y_{ij} - \alpha) \}, \\ Q_{2i}(\boldsymbol{\theta}, \boldsymbol{\omega}' | \hat{\boldsymbol{\theta}}) &= -\frac{1}{2} \log(\sigma_\epsilon^2) - \frac{1}{2\sigma_\epsilon^2} (Y_{0i} - \alpha - \beta X_0)^2. \end{aligned}$$

Under this perturbation scheme, the non-perturbed model follows as $\boldsymbol{\omega}_0 = (1, \dots, 1)^T$. The matrix $\mathbf{\Delta}_{\omega_0}$ has elements, for $j = 1, \dots, m_i, i = 1, \dots, n$, given by

$$\begin{aligned} \left. \frac{\partial Q_{1ij}(\boldsymbol{\theta}, \boldsymbol{\omega} | \hat{\boldsymbol{\theta}})}{\partial \alpha \partial \omega_{ij}} \right|_{\omega_0} &= 0, \quad \left. \frac{\partial Q_{1ij}(\boldsymbol{\theta}, \boldsymbol{\omega} | \hat{\boldsymbol{\theta}})}{\partial \beta \partial \omega_{ij}} \right|_{\omega_0} = 0, \quad \left. \frac{\partial Q_{1ij}(\boldsymbol{\theta}, \boldsymbol{\omega} | \hat{\boldsymbol{\theta}})}{\partial X_0 \partial \omega_{ij}} \right|_{\omega_0} = 0, \\ \left. \frac{\partial Q_{1ij}(\boldsymbol{\theta}, \boldsymbol{\omega} | \hat{\boldsymbol{\theta}})}{\partial \sigma_\epsilon^2 \partial \omega_{ij}} \right|_{\omega_0} &= 0, \quad \left. \frac{\partial Q_{1ij}(\boldsymbol{\theta}, \boldsymbol{\omega} | \hat{\boldsymbol{\theta}})}{\partial \sigma_u^2 \partial \omega_{ij}} \right|_{\omega_0} = \frac{1}{2\sigma_u^2} (\hat{x}_i^2 + X_i^2 - 2\hat{x}_i X_i). \end{aligned}$$

Clearly, the last r columns of $\mathbf{\Delta}_{\omega_0}$, for $i = 1, \dots, r$, have null elements.

4. Simulation study

In order to examine the performance of the Proposed-M, we present some simulation studies. Firstly, we consider both the Proposed-M and Usual-M estimators for comparative purposes.

Secondly, we show the capacity of the methodology to detect atypical data, considering the response variable perturbation scheme. In this study, we use the EM algorithm, where the starting values were used from the explicit ML estimator (see Appendix 2)

4.1. Analysis of the estimators

The aim of this section is to study the performance of the EM estimators of the Proposed-M and verify the impact of erratically considering the Usual-M in the parameter estimates, as well as in the hypothesis tests.

One thousand samples generated from the new approach were considered. In all samples, the value of the parameters α and β were 0.1 and 2, respectively. The range of values for the controlled variable was $[0, 2]$. The fixed values for the controlled variable were $X_1 = 0$ and $X_i = X_{i-1} + 2/(n-1)$, $i = 2, \dots, n$, and the parameter values X_0 were 0.01 (extreme inferior value), 0.8 (near to the central value) and 1.9 (extreme superior value). Consider $\sigma_\epsilon^2 = 0.04$ and $\sigma_u^2 = 0.1$ and 0.01. On the first and second stages, we consider samples of sizes $n = 5, 20$ and 100; $k = 2$ and 20; and $m = 2$ and 10.

The empirical mean bias is given by $\sum_{j=1}^{1000} (\widehat{X}_{0j} - X_0)/1000$ and the empirical mean squared error (MSE) is given by $\sum_{j=1}^{1000} (\widehat{X}_{0j} - X_0)^2/1000$. The mean estimated variance of $\widehat{X}_0$ is given by $\sum_{j=1}^{1000} \widehat{V}(\widehat{X}_{0j})/1000$. The theoretical variance of $\widehat{X}_0$, denoted as $V(\widehat{X}_0)$, is referred to the expression (9) evaluated on the relevant parameter values.

Tables 2 and 3 present the empirical bias and MSE, the estimated variance of X_0 from the proposed and usual model, the theoretical variance $V(\widehat{X}_0)$, the covering percentages (%) and the confidence interval amplitudes A for the parameter X_0 constructed with a 95% confidence level. We observe from Table 2 that the empirical bias and MSE of $\widehat{X}_0$ are small and an increase in the variance size σ_u^2 causes an increase in the empirical bias and MSE of $\widehat{X}_0$ (see Table 3). Moreover, for small sample sizes, independent of X_0 and k , the empirical bias and MSE of $\widehat{X}_0$ decreases when m increases.

We observe in Tables 2 and 3 that for all X_0, n, k and m the estimated variance of the Proposed-M is very close to the theoretical variance, whereas, the estimated variance from the Usual-M does not approach the theoretical value, especially when σ_u^2 is large (see Table 3). Also, analysing

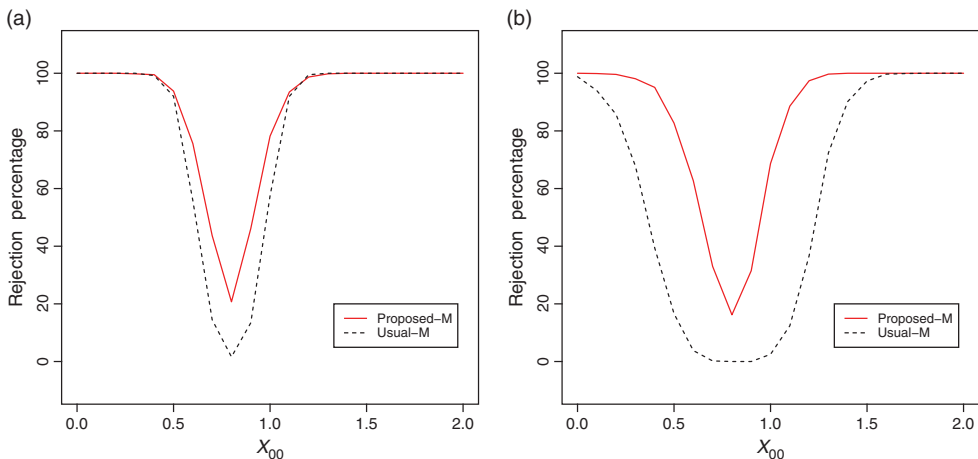


Figure 1. Variation of the percentage of rejection (%) of the hypothesis testing $H_0 : X_0 = X_{00}$ for (a) $\sigma_u^2 = 0.01$ and (b) $\sigma_u^2 = 0.1$.

Table 1. Simulated intensity of the standard solutions at fixed and unknown concentrations.

Intensity					
Fixed concentration					Unknown concentration
$X_1 = 0.0$	$X_2 = 0.5$	$X_3 = 1.0$	$X_4 = 1.5$	$X_5 = 2.0$	
0.1276	1.1816	2.1289	3.5190	4.0180	1.7621
0.0412	0.9863	2.1263	3.2851	4.2804	1.9710
-0.0779	1.2713	1.6594	2.6675	4.2781	1.5624

the amplitude in both Tables 2 and 3, we observe that for all X_0, n, k and m when the Usual-M is erratically adopted, the amplitudes present large values compared with the results of our Proposed-M. In addition an increase in the size of the variance σ_u^2 causes an increase in the amplitude A (see Table 3). We also observe that when m increases, the amplitude A decreases.

Table 2. Empirical bias and mean squared error, the mean estimated variance of $\hat{X}_0$, theoretical variance, covering percentage (%) and amplitude (A) of the intervals with a 95% confidence level for the parameter X_0 , for $\sigma_u^2 = 0.01$.

X_0	n	k	m	Empirical		Mean of $\hat{V}(\hat{X}_0)$		Proposed-M	Usual-M		Proposed-M	
				Bias	MSE	Usual-M	Proposed-M	$V(\hat{X}_0)$	%	A	%	A
0.01	5	2	2	-0.006	0.014	0.010	0.009	0.011	86.40	0.380	81.44	0.352
	5	2	5	-0.006	0.013	0.009	0.007	0.007	87.92	0.366	74.60	0.295
	5	20	2	-0.010	0.010	0.004	0.005	0.006	77.96	0.244	80.24	0.265
	5	20	5	-0.005	0.008	0.002	0.002	0.003	69.26	0.181	70.80	0.188
	20	2	2	-0.002	0.008	0.011	0.007	0.007	97.26	0.401	83.34	0.291
	20	2	5	-0.001	0.007	0.010	0.006	0.006	97.42	0.390	76.12	0.256
	20	20	2	-0.003	0.003	0.002	0.002	0.002	89.22	0.183	89.16	0.184
	20	20	5	-0.001	0.003	0.002	0.001	0.001	84.90	0.151	80.16	0.135
	100	2	2	-0.002	0.006	0.010	0.005	0.005	99.10	0.395	76.60	0.245
	100	2	5	-0.001	0.005	0.010	0.005	0.005	99.02	0.392	73.00	0.232
	100	20	2	0.000	0.001	0.001	0.001	0.001	96.72	0.141	91.94	0.116
	100	20	5	-0.001	0.001	0.001	0.001	0.001	96.90	0.131	88.42	0.100
0.80	5	2	2	-0.002	0.008	0.008	0.007	0.007	90.30	0.330	80.38	0.285
	5	2	5	-0.002	0.008	0.008	0.006	0.006	93.22	0.343	73.84	0.253
	5	20	2	-0.001	0.004	0.002	0.002	0.003	80.84	0.163	82.40	0.174
	5	20	5	-0.002	0.003	0.001	0.001	0.001	76.34	0.135	76.20	0.133
	20	2	2	-0.002	0.006	0.009	0.006	0.006	98.26	0.378	78.12	0.251
	20	2	5	-0.001	0.006	0.009	0.005	0.005	98.56	0.380	72.74	0.235
	20	20	2	-0.002	0.001	0.001	0.001	0.001	92.92	0.136	90.38	0.124
	20	20	5	0.000	0.001	0.001	0.001	0.001	92.14	0.127	85.20	0.104
	100	2	2	0.000	0.005	0.010	0.005	0.005	99.24	0.390	72.62	0.227
	100	2	5	0.000	0.005	0.010	0.005	0.005	99.14	0.389	71.24	0.221
	100	20	2	0.000	0.001	0.001	0.001	0.001	98.62	0.127	93.06	0.096
	100	20	5	0.000	0.001	0.001	0.001	0.001	98.44	0.125	91.44	0.090
1.90	5	2	2	0.004	0.013	0.010	0.009	0.010	87.20	0.372	81.40	0.346
	5	2	5	0.002	0.012	0.009	0.007	0.007	89.14	0.362	74.92	0.286
	5	20	2	0.008	0.009	0.004	0.004	0.006	78.48	0.231	80.14	0.250
	5	20	5	0.005	0.007	0.002	0.002	0.003	69.56	0.173	70.90	0.179
	20	2	2	0.004	0.007	0.011	0.007	0.007	97.28	0.398	82.20	0.286
	20	2	5	0.002	0.007	0.010	0.006	0.006	97.68	0.389	76.82	0.257
	20	20	2	0.002	0.003	0.002	0.002	0.002	88.72	0.175	88.40	0.174
	20	20	5	0.001	0.002	0.001	0.001	0.001	85.64	0.147	81.04	0.130
	100	2	2	0.001	0.005	0.010	0.005	0.005	99.14	0.394	76.54	0.240
	100	2	5	0.001	0.005	0.010	0.005	0.005	99.06	0.392	73.36	0.234
	100	20	2	0.001	0.001	0.001	0.001	0.001	97.04	0.139	91.96	0.113
	100	20	5	0.001	0.001	0.001	0.001	0.001	96.82	0.130	88.84	0.098

Table 3. Empirical bias and mean squared error, the mean estimated variance of $\hat{X}_0$, theoretical variance, covering percentage (%) and amplitude (A) of the intervals with a 95% confidence level for the parameter X_0 , for $\sigma_u^2 = 0.1$.

X_0	n	k	m	Empirical		Mean of $\hat{V}(\hat{X}_0)$		Proposed-M	Usual-M		Proposed-M	
				Bias	MSE	Usual-M	Proposed-M	$V(\hat{X}_0)$	%	A	%	A
0.01	5	2	2	-0.053	0.098	0.061	0.039	0.038	81.50	0.850	71.58	0.660
	5	2	5	-0.049	0.091	0.050	0.020	0.018	81.98	0.793	59.84	0.479
	5	20	2	-0.047	0.087	0.016	0.034	0.033	55.64	0.424	69.66	0.596
	5	20	5	-0.049	0.089	0.011	0.014	0.013	48.06	0.355	52.38	0.395
	20	2	2	-0.015	0.027	0.060	0.016	0.015	99.40	0.939	83.60	0.469
	20	2	5	-0.015	0.026	0.055	0.009	0.009	99.28	0.900	70.66	0.352
	20	20	2	-0.015	0.022	0.011	0.011	0.011	82.78	0.398	82.24	0.396
	20	20	5	-0.015	0.022	0.008	0.005	0.004	75.98	0.341	63.50	0.260
	100	2	2	-0.002	0.009	0.056	0.007	0.007	100.00	0.923	82.58	0.305
	100	2	5	-0.003	0.009	0.055	0.006	0.006	100.00	0.916	73.90	0.262
	100	20	2	-0.004	0.005	0.007	0.003	0.003	98.24	0.326	86.02	0.201
	100	20	5	-0.002	0.005	0.006	0.001	0.001	97.48	0.304	71.76	0.144
0.80	5	2	2	-0.005	0.033	0.043	0.016	0.017	91.30	0.734	75.28	0.443
	5	2	5	-0.009	0.030	0.041	0.010	0.010	92.56	0.723	65.48	0.347
	5	20	2	-0.009	0.030	0.006	0.011	0.012	59.42	0.274	70.14	0.355
	5	20	5	-0.008	0.026	0.005	0.004	0.005	56.44	0.249	54.34	0.237
	20	2	2	-0.001	0.011	0.052	0.008	0.008	99.88	0.876	82.84	0.326
	20	2	5	-0.005	0.011	0.052	0.006	0.006	99.98	0.881	73.28	0.277
	20	20	2	-0.003	0.006	0.006	0.004	0.004	91.94	0.289	83.26	0.226
	20	20	5	-0.002	0.006	0.005	0.002	0.002	91.42	0.282	67.88	0.159
	100	2	2	-0.001	0.006	0.055	0.006	0.006	100.00	0.913	78.42	0.257
	100	2	5	-0.002	0.006	0.055	0.005	0.005	100.00	0.913	72.66	0.237
	100	20	2	-0.001	0.002	0.005	0.001	0.001	99.96	0.291	88.30	0.130
	100	20	5	0.000	0.002	0.005	0.001	0.001	99.94	0.290	80.26	0.106
1.90	5	2	2	0.040	0.087	0.060	0.037	0.034	84.34	0.832	73.76	0.628
	5	2	5	0.040	0.079	0.048	0.017	0.017	82.50	0.775	60.10	0.447
	5	20	2	0.044	0.083	0.017	0.037	0.029	55.34	0.405	69.54	0.568
	5	20	5	0.047	0.076	0.010	0.012	0.012	48.88	0.338	52.90	0.371
	20	2	2	0.016	0.024	0.059	0.015	0.014	99.36	0.931	84.22	0.449
	20	2	5	0.014	0.023	0.055	0.009	0.009	99.48	0.902	70.38	0.339
	20	20	2	0.014	0.019	0.010	0.009	0.009	83.76	0.379	82.38	0.369
	20	20	5	0.013	0.019	0.007	0.004	0.004	79.00	0.332	64.50	0.245
	100	2	2	0.003	0.009	0.056	0.007	0.007	100.00	0.921	82.68	0.298
	100	2	5	0.003	0.009	0.055	0.006	0.006	100.00	0.916	73.10	0.259
	100	20	2	0.003	0.004	0.007	0.002	0.002	98.36	0.319	85.28	0.190
	100	20	5	0.003	0.004	0.006	0.001	0.001	98.42	0.302	73.48	0.138

Figure 1 presents the percentage of rejection (%) for the hypotheses $H_0 : X_0 = X_{00}$, where $X_{00} \in [0, 2]$ and for each X_{00} based on 1000 samples generated from the Proposed-M with $\alpha = 0.1$, $\beta = 2$, $X_0 = 0.8$, $\sigma_e^2 = 0.04$, $n = 5$, $k = 2$, $m = 2$ and (a) $\sigma_u^2 = 0.01$ and (b) $\sigma_u^2 = 0.1$. The hypothesis is tested using the corresponding Wald-type statistic with a significance level of $\alpha = 0.05$. As expected, we also observe that the percent of rejection from the Proposed-M is much greater than the one for the Usual-M. Moreover, when there is an increase in the size of the variance σ_u^2 , it has a much greater impact on the percent of rejection for both models. Overall, these results suggest that inference can be sensitive to model specifications because the operating characteristics of the hypothesis testing and confidence intervals can greatly be affected by the changes.

4.2. Local influence

In order to evaluate the capacity of the methodology to detect atypical data, we simulate a data from the Proposed-M. The value of the parameters α and β were 0.1 and 2, respectively. The range

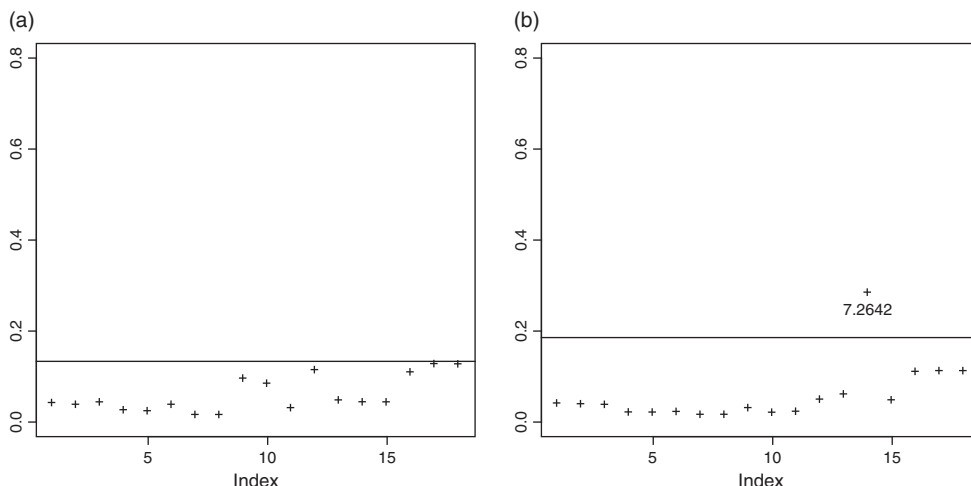


Figure 2. (a) Index plot of $M(0)_j$ for response variable perturbation in a simulated sample. (b) The observation Y_{52} is perturbed.

of values for the controlled variable was $[0, 2]$. The fixed values for the controlled variable were $X_1 = 0$ and $X_i = X_{i-1} + 2/(n - 1)$, $i = 2, \dots, n$, and the parameter value X_0 was 0.8. Consider $\sigma_\epsilon^2 = 0.04$ and $\sigma_u^2 = 0.01$. On the first and second stages, we consider samples of sizes $n = 5$; $k = 3$; and $m = 3$. Table 1 presents the fixed values of concentration of the standard solutions and the corresponding simulated intensities. These data are referred to as the *first stage* of the calibration model. The simulated intensities related to the *second stage* of the calibration model when $X_0 = 0.8$ are also shown in Table 1.

Figure 2(a) depicts the index plots of $M(0)$ for the response perturbation related to Table 1, with the Lee and Xu [14] benchmark computed for $c = 2$. Also, we perturbed the response variable for the maximum value of the variable $\mathbf{Y}$ as follows $Y_{52} = 4.2804 + \sqrt{\mathbf{Y}_5 \mathbf{Y}_5^T}$. In Figure 2(a), we note that the observation Y_{52} is not a influential point and, as expected, in Figure 2(b) such observation is presented as an influential point. This confirms the high sensitivity of M-proposed in the presence of atypical data.

5. Application

In this section, we present some applications using the data supplied by the chemical laboratory of the ‘Instituto de Pesquisas Tecnológicas (IPT) – Brazil’ in order to observe the performance of our Proposed-M, we also fit the Usual-M and both models are compared. The main interest is in estimating the unknown concentration value X_0 of two samples A and B of the barium chemical element.

Table 4 presents the fixed values of concentration of the standard solutions and the corresponding observed intensities for the barium element, which are supplied by the plasma spectrometry method. These data are referred to as the *first stage* of the calibration model. The observed intensities corresponding to the three sample solutions from the samples A and B are also shown in Table 4. These data are referred to as the *second stage* of the calibration model. All estimates were computed considering the intensities divided by 10,000 in order to achieve numerical stability.

Table 5 describes the EM estimate for α , β , X_0 , $V(\hat{X}_0)$, σ_u^2 and the confidence interval amplitude $U(\hat{X}_0)$ to the Proposed-M for samples A and B. The estimates obtained under the Usual-M

Table 4. Concentration (mg/g) and intensity of the standard solutions and the sample solutions A and B of barium element.

Intensity					
Standard solutions				Sample solutions	
$X_1 = 0.1$	$X_2 = 0.2$	$X_3 = 0.5$	$X_4 = 1.06$	A	B
185082.2	373543.2	913829.3	1895804.6	279034.2	2640425.3
184583.0	375166.9	894229.6	1926319.8	279562.1	2661015.0
184906.3	369481.1	911759.2	1886632.8	278462.2	2639452.0

are presented. The estimate for the variance of $\widehat{X}_0$ was computed using the expression (9). The amplitude $U(\widehat{X}_0)$ is given by the product of the squared root of the estimated variance of $\widehat{X}_0$ and 1.96. We can observe that the EM estimate of α and β supplied by the Usual-M is equal to the Proposed-M, this occurs because the expression of the estimators $\hat{\alpha}$ and $\hat{\beta}$ of both models are equal and they only depend on the first stage of the calibration model. Also, we observe that the estimate of the concentration of the sample B is outside the variation range of the standard solution concentrations and the uncertainty of X_0 ($U(\widehat{X}_0)$) from the Proposed-M is slightly greater than the Usual-M estimate.

Testing the hypotheses $H_0 : X_0 = 0.2$ and $H_0 : X_0 = 0.149$ for the sample A and considering 5% of significance level leads to $W = 474.9612(p = 0)$ and $W = 0.01663(p = 0.8974)$, respectively. On the other hand, testing the hypotheses $H_0 : X_0 = 1.6$ and $H_0 : X_0 = 1.48$ for the sample B, by regarding 5% of significance level we have that $W = 562.7771(p = 0)$ and $W = 0.6692(p = 0.4133)$. We observe that when the null hypothesis are the nearest number to the EM estimate the P-values are greater than the significance level, then the result is statistically non-significant. When the null hypotheses are around EM estimates the results are statistically significant.

Figures 3 and 4 present index plots of $M(0)$ for the four perturbation schemes from the sample A and B, respectively. In these figures we conduct a local influence study based on $M(0)$ with interest focussing on θ . In the analysis, we used $c^* = 2$ to compute the benchmark. In Figures 3 and 4, we observe that the intensity 1926319.8, corresponding to the concentration 1.06 from the first step, is detected as influential observation in the weight perturbation and perturbation of the variance σ_ϵ^2 schemes. To see the impact of this observation, we estimated the parameter values considering the Usual-M and Proposed-M for both samples with the complete data set and excluding the

Table 5. Estimates of $\alpha, \beta, X_0, V(\widehat{X}_0)$, the confidence interval amplitude $U(X_0)$ and the relative change from the usual and proposed model for the samples A and B of barium element.

Parameter	Sample A		Sample B	
	Usual-M	Proposed-M	Usual-M	Proposed-M
α	1.139	1.139	1.139	1.139
β	1.786×10^2	1.786×10^2	1.786×10^2	1.786×10^2
X_0	1.499×10^{-1}	1.499×10^{-1}	1.476	1.476
σ_u^2	—	5.749×10^{-7}	—	4.537×10^{-7}
$V(\widehat{X}_0)$	1.260×10^{-5}	1.271×10^{-5}	3.352×10^{-5}	3.394×10^{-5}
$U(\widehat{X}_0)$	6.957×10^{-3}	6.987×10^{-3}	1.135×10^{-2}	1.142×10^{-2}
The relative change				
α	24.1	24.1	23.1	24.1
β	0.7	0.7	0.7	0.7
X_0	0.4	0.4	0.5	0.5
σ_u^2	—	839.8	—	342.1
$V(\widehat{X}_0)$	47.7	57.1	28.2	29.2
$U(\widehat{X}_0)$	27.6	34.5	15.3	15.9

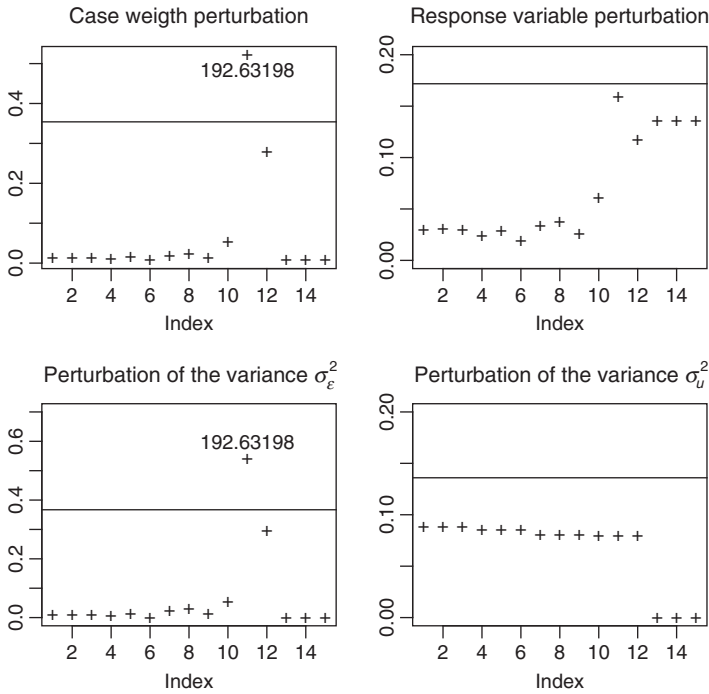


Figure 3. Index plot of $M(0)_j$ for four types of perturbation in sample A.

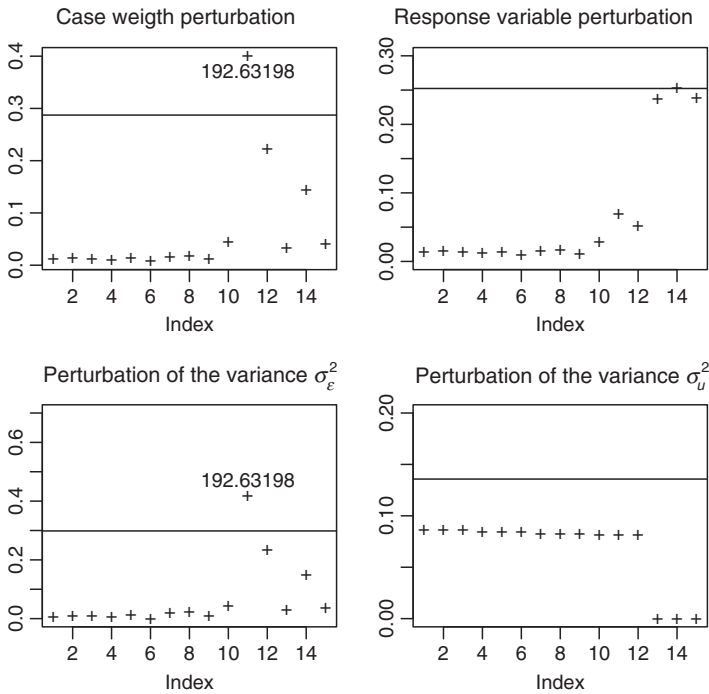


Figure 4. Index plot of $M(0)_j$ for four types of perturbation in sample B.

influential observation. Table 5 presents the relative change $RC = 100|(\hat{\phi} - \hat{\phi}_{(I)})/\hat{\phi}|$, where $\hat{\phi}_{(I)}$ denotes the EM estimator for ϕ after observations belonging to the index set I are removed and $\hat{\phi} = \hat{\alpha}, \hat{\beta}, \hat{X}_0, \hat{\sigma}_u^2, \widehat{V(\hat{X}_0)}$ and $\widehat{U(\hat{X}_0)}$. Observing the relative changes, we note that when we do not consider this influential observation in both samples A and B, the estimates of $\beta, X_0, \sigma_u^2, V(\hat{X}_0)$ and $U(\hat{X}_0)$ from both Usual-M and Proposed-M are strongly affected, being the Proposed-M relative change higher than the Usual-M relative change.

6. Concluding remarks

Simulation studies indicate that when the error variance σ_u^2 increases, the mean estimated variance of $\hat{X}_0$ obtained using the Usual-M moves away from the theoretical value. Usual-M shows a far greater storage confidence interval amplitude than the Proposed-M. This emphasizes the storage uncertainty related to the X_0 concentration. In general, the proposed model is sensible to the presence of the independent measurement error and gives better results in contrast to the Usual-M results. Considering simulated data it is shown that the local influence method on the new approach detect atypical data. In the example above, we observe that one observation is detected as influential data, and the estimation without it differs from the results considering all data. It is observed that despite the fact that the Usual-M, does not consider measurement errors, the amplitude is greater than the one obtained by the new approach. Although we present the procedures in the context of normality, they can be extended to handle controlled calibration models with the normal assumption replaced by other symmetric distributions, such as, the Student-t distribution. We expect to report the results for these more general models in separate papers.

Acknowledgements

The authors thank the editor and two anonymous referees for helpful suggestions and comments. BGBA has been partially financial supported by grants from Brazilian educational agencies: CNPq and FACEPE. VHL was supported in part by grants from FAPESP-Brazil (2010/012465) and CNPq-Brazil (308109/2008-2).

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Appendix 1. Usual-M maximum likelihood estimators

The maximum likelihood estimators of the Usual-M are given by

$$\hat{\alpha} = \bar{Y} - \hat{\beta}\bar{X}, \quad \hat{\beta} = \frac{S_{XY}}{S_{XX}}, \quad \hat{X}_0 = \frac{\bar{Y}_0 - \hat{\alpha}}{\hat{\beta}},$$

$$\sigma_\epsilon^2 = \frac{1}{r + \sum_{i=1}^n m_i} \left[\sum_{i=1}^n \sum_{j=1}^{m_i} (Y_{ij} - \hat{\alpha} - \hat{\beta}X_i)^2 + \sum_{i=1}^r (Y_{0i} - \bar{Y}_0)^2 \right],$$

where

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, \quad \bar{Y} = \frac{1}{N} \sum_{i=1}^n \sum_{j=1}^{m_i} Y_{ij},$$

$$S_{XY} = \frac{1}{N} \sum_{i=1}^n \sum_{j=1}^{m_i} (X_i - \bar{X})(Y_{ij} - \bar{Y}),$$

$$S_{XX} = \frac{1}{N} \sum_{i=1}^n m_i (X_i - \bar{X})^2, \quad \bar{Y}_0 = \frac{1}{r} \sum_{i=1}^r Y_{0i}, \quad N = \sum_{i=1}^n m_i.$$

Appendix 2. Proposed-M maximum likelihood estimators

The maximum likelihood (ML) estimator for the homoscedastic controlled calibration model with replication on both variable (*Proposed-M*) is derived in the following. First, note that the log-likelihood function is given by:

$$l(\alpha, \beta, X_0, \sigma_\epsilon^2, \sigma_u^2) \propto -\frac{\sum_{i=1}^n m_i}{2} \log(\sigma_\epsilon^2 + \beta^2 \sigma_u^2) - \frac{r}{2} \log(\sigma_\epsilon^2)$$

$$- \frac{1}{2} \left[\frac{1}{\sigma_\epsilon^2 + \beta^2 \sigma_u^2} \sum_{i=1}^n \sum_{j=1}^{m_i} (Y_{ij} - \alpha - \beta X_i)^2 \right.$$

$$\left. + \frac{1}{\sigma_\epsilon^2} \sum_{i=1}^r (Y_{0i} - \alpha - \beta X_0)^2 \right]. \quad (\text{A1})$$

Solving $\partial l / \partial \alpha = 0$ and $\partial l / \partial X_0 = 0$ we obtain the ML estimator of α and X_0 which are, respectively, given by

$$\hat{\alpha} = \bar{Y} - \hat{\beta}\bar{X} \quad \text{and} \quad \hat{X}_0 = \frac{\bar{Y}_0 - \hat{\alpha}}{\hat{\beta}}, \quad (\text{A2})$$

where $\bar{Y} = \sum_{i=1}^n \sum_{j=1}^{m_i} Y_{ij} / \sum_{i=1}^n m_i$, $\bar{X} = \sum_{i=1}^n X_i / n$ and $\bar{Y}_0 = \sum_{i=1}^r Y_{0i} / r$.

From Equations (A1) and (A2), it follows that the likelihood for $(\beta, \sigma_\epsilon^2, \sigma_u^2)$ can be written as

$$l(\beta, \sigma_\epsilon^2, \sigma_u^2) \propto -\frac{\sum_{i=1}^n m_i}{2} \log(\sigma_\epsilon^2 + \beta^2 \sigma_u^2) - \frac{k}{2} \log(\sigma_\epsilon^2)$$

$$- \frac{1}{2(\sigma_\epsilon^2 + \beta^2 \sigma_u^2)} \sum_{i=1}^n \sum_{j=1}^{m_i} [(Y_{ij} - \bar{Y}) - \beta(X_i - \bar{X})]^2 - \frac{1}{2\sigma_\epsilon^2} \sum_{i=1}^r (Y_{0i} - \bar{Y}_0)^2.$$

Thus, the ML estimator of β , σ_u^2 and σ_ϵ^2 are given, respectively, by

$$\begin{aligned}\hat{\beta} &= \frac{\sum_{i=1}^n \sum_{j=1}^{m_i} (X_i - \bar{X})(Y_{ij} - \bar{Y})}{\sum_{i=1}^n m_i (X_i - \bar{X})^2}, \\ \widehat{\sigma_u^2} &= \frac{\sum_{i=1}^n \sum_{j=1}^{m_i} \{[(Y_{ij} - \bar{Y}) - \hat{\beta}(X_i - \bar{X})]^2 - \hat{\sigma}_\epsilon^2\}}{\sum_{i=1}^n m_i \hat{\beta}^2}, \\ \widehat{\sigma_\epsilon^2} &= \frac{\sum_{i=1}^r (Y_{0i} - \bar{Y}_0)^2}{r}.\end{aligned}$$